



Assignment: Django REST API

SUBMISSION GUIDELINES

- Create ONE main folder (Example: DjangoRestAPI-YourName).
- Include the complete Django project folder inside it.
- Ensure all required project files are included.
- Convert the folder into ZIP format.
- Upload the ZIP folder on the submission portal.

Zip File Must Contain

- Complete Django project folder (all apps, settings, and files)
- manage.py and db.sqlite3
- All required files: models.py, views.py, serializers.py, permissions.py, filters.py

Problem Statement

Build a RESTful API using Django and Django REST Framework (DRF) for a Blog Post management system. The API should support CRUD operations, user authentication, custom permissions, filtering, searching, ordering, and pagination.

The API should:

1. Allow authenticated users to Create, Read, Update, and Delete blog posts
2. Restrict users to only access and modify their own posts
3. Support filtering posts by creation date
4. Support searching posts by title and content
5. Support ordering by ID
6. Implement pagination (5 posts per page)
7. Use Token/Session-based authentication via DRF

Tech Stack

Component	Technology
Backend Framework	Django 6.0.1
REST Framework	Django REST Framework (DRF)
Filtering	django-filter
Database	SQLite3 (default)
Authentication	Session Auth + DRF Browsable API
Language	Python 3

Project File Structure

```

DjangoRestAPI-YourName/
├── blog/
│   ├── blog/
│   │   ├── __init__.py
│   │   ├── settings.py
│   │   ├── urls.py
│   │   ├── wsgi.py
│   │   └── asgi.py
│   ├── helloworld/
│   │   ├── migrations/
│   │   │   └── __init__.py
│   │   ├── __init__.py
│   │   ├── admin.py
│   │   ├── apps.py
│   │   ├── models.py
│   │   ├── views.py
│   │   ├── serializers.py
│   │   ├── permissions.py
│   │   ├── filters.py
│   │   └── tests.py
│   ├── manage.py
│   └── db.sqlite3

```

Expected Output

Running the project should:

8. User should be able to login using Django authentication
9. API endpoints should work for blog posts
10. Authenticated user should be able to create, view, update, and delete posts
11. User should only access their own posts
12. API should return paginated results (5 posts per page)
13. Filtering, searching, and ordering should work correctly
14. All responses should be in JSON format

If you have any doubt, you can directly contact the Tutedude mentor.